TEAM DOCUMENTATION

Raritone Summer Internship 2026

AI/ML Team – Final Project Documentation

1. Introduction

The AI/ML Team worked on researching and analyzing Artificial Intelligence and Computer Vision technologies for fashion applications. The primary focus of the project was to understand and document the workflow of AI-powered fashion recommendation systems and Virtual Try-On technologies.

The project covered multiple research areas including body detection, pose estimation, cloth segmentation, virtual try-on systems, 3D fashion technologies, recommendation systems, and dataset analysis.

The objective was to explore how these technologies can be integrated to create intelligent, personalized, and interactive fashion shopping experiences.

2. Team Work Approach

To encourage collaboration and active participation, tasks were divided based on individual interests and strengths. Team members selected the areas they were most interested in and contributed through research, documentation, analysis, workflow design, and technology exploration.

This approach allowed the team to explore multiple domains simultaneously while maintaining effective coordination and knowledge sharing.

3. Research Areas Covered

The team collectively researched the following areas:

- Human Body Detection
- Human Pose Estimation
- Body Measurement Analysis
- Cloth Segmentation
- Human Parsing
- 2D Virtual Try-On Systems
- 3D Virtual Try-On Systems
- AI Fashion Recommendation Systems
- AR/VR Fashion Technologies
- Fashion Datasets and AI Models

4. Module 1 – Body Detection System

The team researched body detection technologies used in AI-powered fashion applications.

Activities Performed:

• Studied human body detection workflows.
• Explored body landmark detection methods.
• Researched MediaPipe-based body analysis.
• Analyzed the importance of body detection in virtual try-on systems.

Outcome:

A clear understanding was developed regarding how AI systems identify and analyze human body structures before performing advanced processing tasks.

5. Module 2 – Human Pose Estimation

The team explored pose estimation technologies that identify body keypoints and skeletal structures.

Activities Performed:

• Researched MediaPipe Pose.
• Studied OpenPose architecture.
• Analyzed body landmark detection techniques.
• Investigated skeleton tracking workflows.

Outcome:

The team gained insights into how pose information supports body measurement analysis and garment alignment.

6. Module 3 – Cloth Segmentation & Human Parsing

The team researched segmentation workflows used to separate clothing and body regions from images.

Activities Performed:

• Studied U2Net segmentation models.
• Explored Detectron2-based human parsing.
• Researched image masking techniques.
• Analyzed garment extraction workflows.

Outcome:

The team understood how segmentation contributes to garment fitting accuracy and virtual try-on realism.

7. Module 4 – 2D Virtual Try-On System

The team focused on understanding the workflow of image-based virtual fitting systems.

Activities Performed:

• Studied garment alignment techniques.
• Researched garment warping concepts.

• Explored VITON-HD and HR-VITON architectures.
• Analyzed virtual fitting pipelines.

Outcome:

A complete understanding was gained regarding how garments are digitally aligned and fitted onto user images.

8. Module 5 – 3D Virtual Try-On & AR/VR Research

The team explored advanced virtual fashion technologies and immersive shopping experiences.

Activities Performed:

• Researched 3D avatar generation.
• Studied body mesh reconstruction.
• Explored cloth simulation techniques.
• Analyzed AR and VR fashion applications.
• Evaluated tools such as Blender, Unity, and Three.js.

Outcome:

The team identified future opportunities for immersive fashion experiences through 3D visualization and AR/VR integration.

9. Module 6 – AI Fashion Recommendation System

The team researched intelligent recommendation approaches used in modern fashion platforms.

Activities Performed:

• Studied Content-Based Filtering.
• Explored Collaborative Filtering.
• Analyzed Hybrid Recommendation Systems.
• Researched personalization techniques.

Outcome:

The team gained an understanding of how AI can generate personalized fashion suggestions based on user preferences and behavior.

10. Models Explored

Throughout the internship, the team studied various AI models and frameworks including:

• MediaPipe
• OpenPose
• U2Net
• Detectron2
• VITON-HD
• HR-VITON

These models were analyzed based on their applications, advantages, and relevance to fashion technology systems.

11. Datasets Researched

The team researched multiple datasets commonly used in fashion AI applications.

Datasets Explored:

• DeepFashion
• DeepFashion2
• VITON-HD
• Human Pose Datasets
• Segmentation Datasets

These datasets were studied to understand their role in model training, evaluation, and virtual try-on research.

12. AI Workflow Designed

The team documented a complete AI workflow for intelligent fashion systems.

Workflow:

User Upload
↓
Body Detection
↓
Pose Estimation
↓
Body Segmentation
↓
Measurement Analysis
↓
Garment Mapping
↓
Virtual Try-On
↓
Fashion Recommendation
↓
Final Output

This workflow demonstrates how different AI modules can work together to create personalized shopping experiences.

13. Key Findings

The team identified several important findings:

• Accurate pose detection improves garment fitting.
• Segmentation enhances virtual try-on realism.

• Garment alignment is essential for fitting quality.
• Recommendation systems improve personalization.
• 3D technologies provide immersive user experiences.
• High-quality datasets improve model performance.

14. Challenges Encountered

During the research process, the team identified several challenges:

• Pose variations in user images.
• Segmentation accuracy limitations.
• Garment alignment complexity.
• Dataset diversity requirements.
• Real-time processing constraints.

The team explored various solutions and technologies to address these challenges.

15. Future Recommendations

Based on the research conducted, the team recommends:

• Real-Time Virtual Try-On Systems
• AR-Based Shopping Experiences
• AI Personal Stylist Integration
• Improved Body Measurement Estimation
• Enhanced Recommendation Systems
• Advanced 3D Cloth Simulation
• Mobile-Based Virtual Try-On Solutions

16. Conclusion

The internship provided valuable exposure to Artificial Intelligence, Computer Vision, and Fashion Technology applications. Through collaborative research and documentation, the team gained a comprehensive understanding of body analysis, segmentation, virtual try-on systems, recommendation engines, and AR/VR technologies.

The project successfully demonstrated how AI can be utilized to create intelligent, personalized, and interactive fashion experiences while providing valuable learning opportunities in research, teamwork, documentation, and technology analysis.